

```

# Netflix Data Analysis Project
# Author: DILKHUS KUMAR DAS

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("netflix_data.csv")

# Display first few rows
print("==== Netflix Data Head =====")
print(df.head())

# Check for missing values
print("\n==== Missing Values =====")
print(df.isnull().sum())

# Drop duplicates
df.drop_duplicates(inplace=True)

# Data Overview
print("\n==== Data Info =====")
print(df.info())

# Count of Movies vs TV Shows
type_count = df['type'].value_counts()
print("\n==== Count of Movies vs TV Shows =====")
print(type_count)

plt.figure(figsize=(6, 4))
sns.countplot(data=df, x='type', palette='coolwarm')
plt.title("Movies vs TV Shows on Netflix")
plt.show()

# Top 5 Countries with Most Titles
plt.figure(figsize=(8, 5))
df['country'].value_counts().head(5).plot(kind='bar', color='skyblue')

```

```

df['country'].value_counts().head(5).plot(kind='bar', color='skyblue')
plt.title("Top 5 Countries with Most Netflix Titles")
plt.xlabel("Country")
plt.ylabel("Count")
plt.show()

# Most common genres
plt.figure(figsize=(10, 5))
genres = df['listed_in'].value_counts().head(5)
sns.barplot(x=genres.values, y=genres.index, palette='mako')
plt.title("Top 5 Netflix Genres")
plt.xlabel("Count")
plt.ylabel("Genre")
plt.show()

# Release Year Trend
plt.figure(figsize=(10, 5))
sns.histplot(df['release_year'], bins=10, kde=True)
plt.title("Number of Releases per Year")
plt.xlabel("Release Year")
plt.ylabel("Count")
plt.show()

# Rating Distribution
plt.figure(figsize=(8, 5))
sns.countplot(data=df, y='rating', order=df['rating'].value_counts().index, palette='Set2')
plt.title("Distribution of Content Ratings")
plt.xlabel("Count")
plt.ylabel("Rating")
plt.show()

print("\n==== Insights =====")
print("1 Movies dominate over TV Shows.")
print("2 Most Netflix titles come from the United States and India.")
print("3 Popular genres include Drama, Action, and Sci-Fi.")
print("4 Netflix content production has increased rapidly after 2015.")
print("5 The majority of content is rated for mature audiences (TV-MA or R).")

```


==== Netflix Data Head ====

	show_id	type	title	director	cast \
0	s1	Movie	Extraction	Sam Hargrave	Chris Hemsworth
1	s2	TV Show	Money Heist	Álex Pina	Úrsula Corberó
2	s3	Movie	Bird Box	Susanne Bier	Sandra Bullock
3	s4	Movie	The Irishman	Martin Scorsese	Robert De Niro
4	s5	TV Show	Stranger Things	The Duffer Brothers	Millie Bobby Brown

	country	date_added	release_year	rating	duration \
0	India	April 24, 2020	2020	R	117 min
1	Spain	December 20, 2017	2017	TV-MA	5 Seasons
2	United States	December 21, 2018	2018	R	124 min
3	United States	November 27, 2019	2019	R	209 min
4	United States	July 15, 2016	2016	TV-14	4 Seasons

	listed_in	description
0	Action & Adventure	Tyler Rake is a fearless black market mercenar...
1	Crime TV Shows	An unusual group of robbers attempt to carry o...
2	Thrillers	Five years after an ominous unseen presence dr...
3	Dramas	An aging hitman looks back on his career and t...
4	Sci-Fi & Fantasy	A group of young friends witness supernatural ...

==== Missing Values ====

show_id	0
type	0
title	0
director	0
cast	1
country	0
date_added	0
release_year	0
rating	0
duration	0
listed_in	0
description	0

dtype: int64

==== Data Info ====

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -

```